

Product Requirements Document (PRD) v2.0

CloudWise AI

Version: 2.0

Status: Active Development

Last Updated: June 2026

1. Product Overview

CloudWise AI is a cloud cost optimization and infrastructure intelligence platform designed to help organizations understand, monitor, and optimize their AWS cloud spending.

The platform connects to either a real AWS environment or a LocalStack development environment, discovers cloud resources, estimates costs, identifies optimization opportunities, and provides actionable recommendations through a modern analytics dashboard.

CloudWise AI aims to simplify FinOps practices for startups, developers, and small-to-medium businesses by making cloud cost management accessible and understandable.

2. Problem Statement

Many organizations struggle with:

- Rising AWS bills
- Unused or underutilized cloud resources
- Lack of visibility into infrastructure costs
- Difficulty identifying optimization opportunities
- Limited FinOps expertise

As cloud infrastructure grows, manual cost analysis becomes inefficient and error-prone.

Organizations need a centralized platform that automatically discovers resources, analyzes usage patterns, estimates costs, and recommends savings opportunities.

3. Product Vision

To become an intelligent cloud optimization platform that continuously monitors cloud infrastructure, explains spending patterns, predicts future costs, and provides AI-powered recommendations that reduce cloud waste.

4. Goals

Business Goals

- Reduce cloud waste
 - Improve infrastructure visibility
 - Simplify cloud optimization
 - Increase operational efficiency
 - Provide FinOps insights without requiring FinOps expertise
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User Goals

Users should be able to:

- Connect AWS accounts
 - Connect LocalStack environments
 - Scan cloud resources
 - View infrastructure inventory
 - Monitor cloud spending
 - Identify optimization opportunities
 - Understand resource utilization
 - Forecast future spending
 - Generate optimization reports
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5. Target Users

Primary Users

Startup Founders

Need visibility into cloud costs without dedicated DevOps teams.

Developers

Need resource visibility and optimization recommendations.

DevOps Engineers

Need faster cloud audits and cost analysis.

Cloud Architects

Need infrastructure inventory and optimization insights.

6. User Personas

Persona 1

Name: Startup Founder

Goals:

- Reduce monthly AWS bills
- Understand spending trends

Pain Points:

- Limited cloud expertise
 - Unexpected cloud costs
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Persona 2

Name: DevOps Engineer

Goals:

- Identify waste
- Optimize infrastructure

Pain Points:

- Manual resource reviews
 - Time-consuming audits
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7. Core Features

Authentication

Features

- User Registration
- User Login
- JWT Authentication
- Forgot Password
- Password Reset
- Protected Routes

Priority: P0

Cloud Connection

AWS Mode

Features:

- AWS Access Key authentication
- AWS Secret Key authentication
- STS validation
- Region selection

Priority: P0

LocalStack Mode

Features:

- LocalStack toggle
- Automatic endpoint switching
- Local cloud simulation

Endpoint:

`http://localhost:4566`

Priority: P0

Resource Discovery

Current Services:

EC2

Collected Data:

- Instance ID
 - Instance Type
 - Status
 - Region
 - Tags
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EBS

Collected Data:

- Volume ID
- Size
- State
- Attachments
- Encryption

Priority: P0

Future Services:

- S3
- RDS
- Lambda

Priority: P1

Resource Inventory

Features:

- Resource listing
- Search
- Filtering
- Status visibility

Priority: P0

Cost Intelligence

Features:

- Monthly spend tracking
- Daily spend tracking
- Cost breakdown analysis
- Trend visualization

Priority: P0

Recommendation Engine

Current Rules:

EC2

- Idle Instance Detection
- Underutilized Instance Detection

EBS

- Unattached Volume Detection

Outputs:

- Recommendation
- Priority Level
- Estimated Savings

Priority: P0

Dashboard Analytics

Features:

KPI Cards

- Total Resources
- Monthly Spend
- Potential Savings
- Health Score

Charts

- Cost Trend
- Resource Distribution

Summary Panels

- Recommendations
- Infrastructure Overview

Priority: P0

Forecasting Engine

Features:

- 7-Day Forecast
- 30-Day Forecast
- 90-Day Forecast

Priority: P1

AI Copilot

Features:

- Explain recommendations
- Explain spending patterns
- Explain forecasts
- Answer infrastructure questions

Example:

"Why is my EC2 spend increasing?"

Priority: P1

Report Generation

Features:

- Executive Summary
- Optimization Report

- Cost Analysis Report

Export Formats:

- PDF
- CSV

Priority: P1

8. User Journey

AWS Flow

Signup



Login



Connect AWS



Validate Credentials



Scan Resources



Discover Infrastructure



Calculate Costs



Generate Recommendations



View Dashboard



Download Reports

LocalStack Flow

Signup



Login



Enable LocalStack



Connect Local Environment



Discover Mock Resources



Generate Analytics



View Dashboard

9. Functional Requirements

FR-1 Authentication

System shall allow users to create accounts.

FR-2 Login

System shall authenticate users using JWT.

FR-3 Password Recovery

System shall allow password reset via forgot password workflow.

FR-4 AWS Validation

System shall validate AWS credentials through STS.

FR-5 Resource Discovery

System shall discover EC2 and EBS resources.

FR-6 Cost Analysis

System shall calculate estimated cloud costs.

FR-7 Recommendations

System shall generate optimization recommendations.

FR-8 Dashboard

System shall display cloud analytics.

FR-9 Forecasting

System shall estimate future cloud spending.

FR-10 Reports

System shall generate downloadable reports.

10. Non-Functional Requirements

Performance:

- Dashboard load < 3 seconds
- Resource scan < 60 seconds

Reliability:

- 99% successful scans

Scalability:

- Support multiple cloud accounts

Security:

- Encrypted credentials
- JWT authentication
- Secure API communication

Usability:

- Minimal learning curve
 - Modern responsive UI
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11. Success Metrics

Product success will be measured by:

- Successful cloud account connections
 - Successful resource scans
 - Number of recommendations generated
 - Estimated savings identified
 - Dashboard engagement
 - Report generation usage
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12. MVP Scope

Included:

- Authentication
- Forgot Password
- AWS Integration
- LocalStack Integration
- EC2 Discovery
- EBS Discovery
- Resource Inventory
- Dashboard
- Recommendations

Excluded:

- Azure Support
 - GCP Support
 - Team Collaboration
 - RBAC
 - Multi-Tenant Management
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13. Future Roadmap

Version 2.1

- Forecasting Engine
 - Health Score Improvements
 - Report Generation
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Version 3.0

- AI Copilot
 - S3 Discovery
 - RDS Discovery
 - Lambda Discovery
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Version 4.0

- Azure Integration
- GCP Integration

- Multi-Cloud Optimization
 - Team Collaboration
 - Role-Based Access Control
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Deployment

Frontend:

Vercel

Backend:

Render

Database:

PostgreSQL

Development Environment:

LocalStack

Cloud Provider:

AWS

Status: Approved v2.0